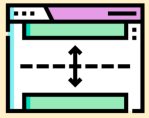
Visual Abstract — Lightweight, Explainable Multi-Class Detection and Recovery of Format- and Substitution-Based Text Steganography

Problem

Covert text channels
Zero-width characters, double spaces and synonym swaps hide payloads in normal-looking text



Detection gap
In-domain accuracy hides false alarms, seed spread and domain shift



Heavy detectors
Neural detectors need millions of parameters and give no per-alert explanation



Robustness unknown
Payload size, normalisation and unseen embedders are rarely stress-tested



Method

Leakage-free benchmark
IMDB train/test; WikiText-2 and SMS Spam external; split grouped by cover text



Lightweight detector
4 classes: Clean, ZeroWidth, Whitespace, Synonym; 17 features + LightGBM-Lite



Strong baselines
Rule, TextCNN, ALBERT, RoBERTa, Qwen-LoRA; multi-seed runs, cluster-bootstrap CIs



Explain and reveal
SHAP (global + per alert), then decode the payload with the matching decoder



Results

Easy in-domain
IMDB macro-F1: Lite 0.97, ALBERT 0.996; a simple 3-rule baseline already reaches 0.96



Stable and calibrated
Lite: seed std ≤ 0.003; external FPR 1.7% / 0.2% at a 1%-FPR threshold (ALBERT 43% / 77%)



Lightweight
Lite: ≈0.4 ms, ≈10k tree nodes; ALBERT: ≈11 ms, 11.7M parameters



Honest limits
Unseen format encodings: 0.00 recall; normalisation destroys the payloads



Impact

Deployable
CPU-only detect, explain and reveal pipeline with a Gradio app



Explainable alerts
Each alert carries SHAP reasons and, when possible, the recovered payload



Reproducibility
Fixed seeds, split IDs, code and models on GitHub / Zenodo / Hugging Face



Future roadmap
Generation-based stego, real-world benchmarks, multi-embedder training

